

GENERAL SPECIFICATION G4.3

Lifting Equipment

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G4.3 LIFTING EQUIPMENT

G4.3.1 General

- (a) All lifting equipment shall be provided complete with drive units, access platforms, ladders, runway beam, corbels, supports, plinths, painting, labelling, field control panel, emergency stop button, cabling, etc., with all necessary instruments required to control the lifting operations and all other necessary accessories and ancillaries required to form a complete and functional system.
- (b) All lifting equipment shall be registered with the Factory Inspectorate under the Ministry of Manpower (MOM), Singapore and copies of registration certificate (from MOM) for the crane/lifting equipment must be handed to the S.O..
- (c) The Contractor shall be responsible to conduct the periodic tests on the lifting equipment as stipulated in the Workplace Safety and Health regulations and also for maintaining the lifting equipment registration and re-registration processes with MOM, including all fees to be paid until the end of defects liability period (DLP).
- (d) All lifting equipment provided shall bear a permanent inscription on each side legible from ground floor level stating the safe working load and the registration number assigned by the Factory Inspectorate.
- (e) The Contractor shall provide the test loads, any instrumentation required and the level of supervision as required by the concerned authorities. The test load shall be removed from site by the Contractor after successful tests have been carried out.
- (f) The Contractor shall make his own assessment, by calculation or approved load testing equipment, of the weight of main components to be lifted and, prior to lifting, shall satisfy the S.O. that the safe working loads (SWL) of lifting equipment are not exceeded.
- (g) All relevant drawings and calculations of the lifting equipment shall be with Professional (P.E.) endorsement and submitted to the S.O. for approval before any fabrication and installation is carried out.
- (h) Lifting gear shall be designed to allow a straight lift to be achieved.
- (i) The “Safe Working Load” shall be marked conspicuously and clearly on all lifting equipment.
- (j) The Contractor shall engage a P.E. to design the crane system and its associated equipment including any structural design. The Contractor shall also be responsible for any necessary authority submission and approval. The Contractor shall submit application to MOM and engage an MOM approved personnel to test the lifting equipment and certify that it is safe for use. The P.E. certification and registration with MOM shall be valid until the end of the defects liability period (DLP) for the Works.
- (k) Laboratories accredited by the Singapore Accreditation Council (SAC) under the SINGLAS scheme shall carry out all mechanical and non-destructive testing for all lifting equipment, as required by the Chief Inspector of Factories, Ministry of Manpower. Only test reports issued by SAC-SINGLAS accredited laboratories shall be recognised.

G4.3.2 Reference Standards

- (a) Applicable standards governing lifting devices include but are not limited to the following:

- (i) American National Standards Institute (ANSI):

Standard	Subject
B30.2	Overhead and Gantry Cranes (Top Running, Single or Multiple Girder)
B30.10	Hooks
B30.11	Monorails and Underhung Cranes
B30.16	Overhead Hoists (Underhung)
B30.17	Overhead and Gantry Cranes (Top Running, Single Girder)
HST 1M	Performance Standard for Electric Chain Hoists
HST 2M	Performance Standard for Hand Chain Manually Operated Chain Hoists
HST 4M	Performance Standard for Overhead Electric Wire Rope Hoists
MH27.1	Underhung Cranes and Monorail Systems

- (ii) British Standards Institute (BSI):

Standard	Subject
BS EN 13001	Cranes. General design. General principles and requirements
BS EN 15011	Cranes, Bridge, and Gantry Cranes
BS EN 12385-1:2002+A1:2008	Steel wire ropes. Safety. General requirements
BS EN 3506	Mechanical properties of corrosion-resistant stainless-steel fasteners. Bolts, screws and studs
ISO 11660	Cranes. Access, guards, and restraints. Jib cranes
BS EN 14985	Cranes slewing jib cranes

- (iii) Crane Manufacturer's Association of America (CMAA):

- 1) 70, Electric Overhead Travelling Cranes; and
- 2) 74, Top Running & Under Running Single Girder. Electric Overhead Travelling Cranes.

- (iv) Factories Act (Chapter 104), Singapore.

- (v) International Electro technical Commission (IEC): 60529, Degrees of Protection Provided by Enclosures (IP Code).

- (vi) National Electrical Manufacturer's Association (NEMA): MG 1, Motors and Generators.
- (vii) Singapore Productivity and Standards Board (PSB):

Standard	Subject
CP 5	Electrical Installations
CP 16	Earthing;
CP 27	Factory Layout Safety, Health, and Welfare Considerations
SS 497	Electric Overhead Travelling Cranes

- (viii) The Contractor and the sub-contracted suppliers/manufacturers shall be responsible to use the latest versions of standards applicable at the time of project execution.

G4.3.3 Submittals

- (a) Contractor shall submit the following as minimum:
 - (i) Make, model and weight of each equipment assembly;
 - (ii) Complete catalogue information, descriptive literature, materials of construction, and specifications on hoist, wheels, gears and bearing, long travel speeds, cross travel speeds, hoisting speeds, hoist motor, end-truck motor, cross-travel motor, auxiliary motor (where applicable) and assemblies, hook, brakes and accessories;
 - (iii) Structural design calculations for crane and associated structure, jib crane mast, boom, support system, connection to main structure, and calculations of deflection and loads on building. Designs and all calculations shall be endorsed by the Contractor's P.E. registered in the relevant discipline;
 - (iv) Detail Shop Drawings of crane system, jib crane mast, boom and their attachments to building concrete;
 - (v) Layout drawings showing crane traverse span, long travel extents, lowest and highest hoisting hook locations, area coverage, required headroom space, installation/erection and maintenance space requirements;
 - (vi) Factory finish system;
 - (vii) Operation and Maintenance Manuals; and
 - (viii) Registration Certificate with Chief Inspector of Factories.
- (b) Refer to electrical General Specifications for motor submittals.
- (c) The Contractor shall also submit the following documents as minimum requirement for the Quality Control Submittals:
 - (i) Factory Functional Test Report;
 - (ii) Factory and field test procedures;

- (iii) Manufacturer's Certification of Compliance that the factory finish system is identical to the requirements specified herein;
 - (iv) Special shipping, storage, protection and handling instructions;
 - (v) Manufacturer's printed installation instructions;
 - (vi) Manufacturer's Certificate of Proper Installation, including Site Functional Test Report; and
 - (vii) Statement from Manufacturer confirming 5 years of experience with similar equipment configuration and capacity required for the works.
- (d) Suggested spare parts list to maintain the equipment in service for a period of 1 to 5 years. Include a list of special tools required for checking, testing, parts replacement, and maintenance with current price information.
- (e) List special tools and materials supplied with equipment for use prior to and during commissioning and for future maintenance.
- (f) Provide warranty with the Board named as beneficiary for complete equipment including accessories and control system as specified under this contract.

G4.3.4 Quality Assurance

- (a) Qualifications
- (i) Manufacturer shall have a minimum of 5 years' experience manufacturing, providing, installing and commissioning similar equipment as required for the Works.
 - (ii) Reference list of installations of similar equipment that has been in operation for the past 5 years.

G4.3.5 Electric Overhead Travelling Cranes

- (a) General
- (i) Cranes shall be designed in accordance with BS EN 13001 and BS EN 15011:2011+A1:2014.
 - (ii) The term crane shall be deemed to include gantry rails, platform with hand rails for maintenance purposes, access ladder, down shop conductors, end stops, holding down bolts and all other items required for the complete installed operational crane system.
 - (iii) The longitudinal station travel, the transverse travel and the hoist shall be electrically operated. The hoist must be capable of reaching every part of the area served and must be able to handle all the corresponding installed equipment.
 - (iv) Operation shall be from ground floor level by bridge-mounted pendant push-button controls or remote control as specified.
 - (v) All the movements of the crane shall be geared so that one man of average stature and strength can move and control the fully laden crane with accuracy.

- (vi) Controls shall be mechanically and electrically interlocked to prevent inadvertent operation of opposing motions. The pendant control voltage shall not exceed 50 V AC.
- (vii) The pendant shall be supported independently of the electric cable and shall be arranged for extending when necessary. Pendant wall mounting hook shall be provided and installed as required by S.O..
- (viii) The crane shall be fitted with limit switches to prevent excess travel, over-hoisting and over-lowering of the crane hook.
- (ix) Electro-mechanical fail-safe brakes shall be incorporated on all power-operated motions.
- (x) The crane shall be fitted with a load-limiting device to stop the lifting operation when the load exceeds the pre-set limit.
- (xi) The load sensor shall be of the strain gauge bridge transducer type calibrated at the factory.
- (xii) Crane hooks shall be fitted with safety catches and the hook block shall incorporate fully guarded rope sheaves.
- (xiii) Down shop conductors shall be of the fully insulated shrouded bus bar type. The current collectors shall have renewable contact pieces. Festoon cables may be used for the cross travel.
- (xiv) A crane power supply isolating switch, lockable in the off position and incorporating a supply on indicator light, shall be provided at the bottom of the access ladder.
- (xv) A second isolating switch shall be provided at the control cubicle located on the crane platform.
- (xvi) All motors shall be of the quick reversing type.
- (xvii) All movements shall be electrically operated and shall function with the hook loaded. Facilities shall be provided for the accurate location of the hook by means of "inching" the cross travel and down shop travel motions.
- (xviii) All the motions shall be suitable for the dismantling and accurate re-assembly of the equipment parts. The range of speeds shall be designed to give ease of handling and maintenance of equipment without danger to the people, staff or pumping station, installation etc.
- (xix) Sufficient slings, ropes, shackles, and lifting spreader beams shall be supplied to handle all items of plant served by the crane. They shall be labelled or marked with the Safe Working Load (SWL) and the purpose for which they are intended.
- (xx) All necessary contactors, control cubicles and protection equipment necessary to operate the crane and provide adequate electrical protection against overload, phase and earth fault and fail-safe protection in the event of an interruption in the power supplies shall be provided.

- (xxi) All electrical equipment shall be fully tropicalized where necessary. Motors and switchgear shall be provided with anti-condensation heaters which shall be energised when the crane is at rest and suitable warning labels shall be fitted.
- (b) Crane Accessories
 - (i) Hoisting Unit
 - 1) The hoisting unit shall be of an electric driven wire rope type. It shall be capable of safe operation with optimum acceleration characteristics. Importance shall be given to minimum maintenance.
 - 2) The design shall incorporate a normal hoisting speed and a much slower creeping speed. Adequate interlocking and control shall be provided to prevent over travel both upwards and downwards.
 - 3) The crane hoist shall be capable of lifting and traversing in two directions and to hoist a maximum load as specified.
 - (i) Cross Travel Unit
 - 1) The above unit, namely the hoist trolley, shall have an electric drive. Safe operation and optimum acceleration characteristics of the cross-travel unit shall be ensured through the use of individual drive units incorporating self-braking motors. Hoist trolley shall operate through the entire distance along the crane girder
 - 2) Importance shall be given to the presence of features allowing for minimum maintenance.
 - (i) Long Travel
 - 1) The long travel unit shall have an electric drive. Safe operation and optimum acceleration characteristics of the long travel unit shall be ensured through the use of individual drive units incorporating self-braking motors.
 - 2) Importance shall be given to the presence of features allowing for minimum maintenance.
 - 3) The drive shall be of the direct torque transmission type. Transmission to the travel wheel shall be by splines through reduction gears and with the gear wheels running in oil-bath and with all shafts on anti-friction bearings.
 - 4) The long travelling wheels shall be of double flange special spheroidal-graphite cast iron type.
 - (ii) Hoisting Rope
 - 1) The rope shall be of the best plough steel, of twist-free type and shall comply with BS EN 12385-1:2002+A1:2008.
 - 2) Its breaking strength shall be not less than six times the safe working load (SWL). Sufficient length of this rope shall be provided.

- (iii) Rope Drum
 - 1) The rope drum shall be rolled, welded and precision machined from high grade steel and shall be flanged at both ends. It shall be driven through an internal gear rim and supported by large diameter bearings. The drum shall have specially machined grooves.
 - 2) The size of the drum grooves shall be such as to accommodate within 1 layer the length of rope required for the specified lift. Furthermore, it should have no fewer than 3 dosed turns at each anchored and a spare groove at the other end.
- (iv) Rope guide: The rope guide shall consist of a pressure ring and a threaded ring smoothly guiding the rope off the drum and operating the limit switch in the event of hook over travel.
- (v) Hook Block: The hook block shall be of the fully-enclosed type conforming to BS EN 1677 Part 5. The material of the hook shall be forged steel. The hook shall be fully swivelling and shall be mounted on a ball or roller bearing. A protective skirt shall be provided to enclose the bearing. A safety catch shall also be provided. The block must be arranged for lowering down to the floor level. The shape of the block shall be designed to avoid the catching of the block when lifting through floors and openings.
- (vi) Hoist rope pulleys: The above pulleys are to be of cast steel fitted with roller bearings and arranged for grease lubrication.
- (vii) Brakes
 - 1) All motors shall be equipped with conical rotor electric mechanical brakes. The springs for the brakes shall be of the compressive type and shall not be stressed in excess of 80% of the torsional elastic limit of the material.
 - 2) All brake linings shall be effectively and permanently secured. Provisions shall be made for simple and easily accessible means of adjustment due to wearing of the lining.
 - 3) All brakes shall be protected from oil, grease, dust and any other adverse effects.
 - 4) The brake for the hoisting unit shall be the automatic type. When the circuit is cut-off or fails, it shall be able to arrest the motion and hold at rest any load up to and including the test load at any position of the lift. It shall be able to safely control the lowering of any load up to and including the test load. The brake torque shall not be less than 250% of the rated motor torque.
 - 5) The brakes for the traversing and travelling units shall be able to bring the crane safely to rest at any load up to and including the test load and shall be of the automatic type.

- (viii) Gears, Bearings and Lubrication
- 1) All gears are to be mounted on anti-friction bearings and totally enclosed within an oil tight housing. They are to be of high quality gear steel and are to possess accurately machined teeth. The gear teeth are in addition required to be hardened and highly resistant to wear and fracture.
 - 2) The bearings are to be of the anti-friction type and required to be permanently lubricated. The bearings shall be designed for a minimum service life of 5 years.
 - 3) Re-lubrication, if necessary, shall be made possible through easily accessible grease nipples. The latter shall be of uniform type and readily accessible. Special attention is to be given to the lubricating of points not readily accessible.
- (ix) Crane Runway Rails
- 1) Contractor shall design, supply and install the crane runway rails. Rails shall be constructed from the steel universal I-beam.
 - 2) The sizing of the steel universal I-beam shall take into account of the total load of the overhead travelling crane and the rated working load with the appropriate safety factor for an efficient operation of the overhead travelling crane.
 - 3) The rails for the portal cranes shall be constructed of square bar of steel suitable for the portal crane application. The rails shall be welded to the levelling plates which extend throughout the whole length of the rails.
- (x) Transverse Crane Beam
- 1) Transverse Crane Bridge shall be a single box girder mounted on the top of double channel end carriages for crane with a safe working load of less than 10 tons.
 - 2) Transverse Crane Bridge shall be a double box girder of double rail crab for crane with a safe working load of 10 tons and above.
 - 3) The box section shall comprise two rolled steel joints with the flanges butting and continuously welded to provide a horizontally stiff girder.
- (xi) Crane Rail Support Joints
- 1) The rolled steel support members to which the crane rails are to be fixed shall be provided. The joints are to be of approved dimensions and type capable of adequately supporting the fully loaded crane at all points and during travel without significant deflection.
 - 2) The members are to be securely fixed to the supporting corbels from reinforced concrete columns provided by the main Contractor, by purpose made shoes and deep stainless anchor bolts. The fixings shall be of an approved design such that the members are simply supported on the columns and impart no fixing moments.
 - 3) End stops shall be installed to arrest the crane.

- (xii) Crane Rails
 - 1) The crane rails complete with fish plates, holding down bolts and stops etc. are to be supplied and fixed complete. The rails and fittings are to be of heavy gauge flat bottom type.
- (xiii) Bolts, Nuts and Washers
 - 1) Bolts, nuts and washers shall be of stainless steel and shall conform to BS EN 3506 - Specification for corrosion-resistant stainless steel fasteners.
- (c) Control of Crane
 - (i) General
 - 1) The complete electrical isolation of the crane shall be controlled from the main electrical control panel and such isolation shall have the effect of stopping all hoist movements.
 - 2) The crane shall be controlled by low voltage operation via reversing magnetic contactors from a pendant push-button station as far as applicable. The control voltage, preferably 42V, shall be supplied by a transformer, the primary and secondary windings of which are to be protected by special fuse links.
 - 3) Adequate control will be levied through the use of suitably placed limit switches and brake mechanisms. Sufficient interlocking will also be provided to ensure a fail-safe unit.
 - (ii) Pendant Push-Button System
 - 1) If crane control is specified to be controlled from a push-button station, it shall be of modern, practical and handy design and suitable for floor operation.
 - 2) The push-button station shall be suspended from the crane bridge and be mobile along one side of the bridge through the use of trailing cables supported by ball bearings trolleys or sliders running smoothly on a rigid light aluminium track adjacent to the bridge.
 - 3) The control pendant shall be lockable to prevent misuse and shall be provided with a special flexible hose to protect the control cable against tensile stresses and possible mechanical damage.
 - 4) The weight of the control pendant shall be supported independently of the electric cable. If the control pendant is made of metal, it shall be effectively earthed.
 - 5) The push-buttons shall be housed in a sturdy, dust-proof box of approved material so designed as to give the required mechanical or electrical protection to all operating parts.
 - 6) The control pendant shall be capable of withstanding damage.

- 7) Each crane motor will be operated by two (2) push-buttons, one for each direction of motion. Each push-button is to be of the double pressure type for dual speed operation. All the buttons shall be of the re-setting type to allow the hoist to be ready for subsequent operation by any other control button.
 - 8) The push-button station for the portal crane shall be of weatherproof construction with a protection of IP 65.
 - 9) The pendant control shall be positioned 1.3 m above the operation and control floor level or as per specified.
 - 10) The pendant and cable shall be supported by a flexible non-ferrous or galvanized steel wire rope secured by splices and shackles.
- (iii) Limit Switches
- 1) Long travel and cross travel limit switches with over-riding facilities shall be incorporated and suitably arranged in the crane unit.
 - 2) Limit switches required to cut-off the raising and lowering motions when the hook block reaches its top and bottom maximum limits shall be provided.
 - 3) Position limit switch shall be provided where more than one trolley is operated on the same bridge to prevent the overshoot of the crane and collision.
- (iv) Load Limiting Device
- 1) Each crane shall be equipped with a load limiting device which can measure the mechanical load being hoisted to cease lifting operation (and maintain the load in place) when the load exceeds the pre-set limit. The load sensor shall be of the strain gauge bridge transducer type calibrated at the factory.
- (d) Electrical Works
- (i) General
- 1) Contractor shall supply and install all the electrical works necessary for the proper functioning of the electric overhead and portal crane systems.
 - 2) All the electrical works shall comply with the specifications of the distribution system in General Specification - Electrical Works.
- (ii) Motor Control Centre (MCC)
- 1) MCC shall be complete with the following components:
 - Motor starters;
 - Moulded Case Circuit Breakers (MCCBs)/ Miniature Circuit Breakers (MCBs);
 - Main isolator; and
 - Name plate.

(iii) Power Supply and Crane Wiring

- 1) The crane shall operate on a 3 phase, 415 Volts, 50 Hz supply. Supply shall be 60 volts fed from a transformer with an earthed centre tap secondary.
- 2) Generally, the power supply to the cranes shall be through totally enclosed copper bus-ducts suitable for safe mobile power feeding to the cranes except the portal crane.
- 3) The supply across the gantry beams shall be made by means of heavy duty type multi core PVC cable running along a collector channel and loading onto a spring loaded drum. The supply along the length of the station must be completely insulated in a PVC housing and protected against any accidental contact. The wiring collector and pick-up gear must be arranged for ease of maintenance from the crane gantry.
- 4) A red warning lamp shall be provided at each end of the track and shall be illuminated when the power is on.
- 5) The controller shall be fixed to the crane and arrange for easy maintenance.

(iv) Motors

- 1) The motors shall be squirrel cage induction motors with adequate surface ventilation for intermittent duty and comply with requirements in the electrical General Specification on motors.
- 2) The motor ratings shall conform to the relevant FEM design rules. An allowance of at least 25 % shall be included. Insulation shall be fully tropicalized and be preferably categorised in Class F.
- 3) The type of enclosure for the motor shall be dust and damp proof and fitted with anti-condensation heater as minimum. Motor enclosure shall also conform with the appropriate IEC rules and the requirements as specified in electrical General Specification on motors.

(v) Earthing

- 1) The crane structure, motor frames and metal cases of all electrical equipment including metal conduit or cables guards shall be effectively earthed.

G4.3.6 Manually Operated Overhead Cranes

- (a) Cranes shall be designed in accordance with BS EN 13001 and BS EN 15011.
- (b) The term crane shall be deemed to include gantry rails, end stops, holding down bolts and all other items required for the fully installed operational crane system. All access ladders and platforms necessary to carry out maintenance and repairs shall also be provided.
- (c) Sufficient slings, ropes, shackles, and lifting beams shall be supplied to handle all items of Plant served by the crane. They shall be labelled or marked with their safe working load and the purpose for which they are intended.

G4.3.7 Jib Cranes and Davit Cranes

- (a) The lifting capacity of the jib or davit cranes shall be according to the Particular Specifications and the Contractor shall verify the loading capacity for the selected lifting device and submit the loading schedule for S.O. approval.
- (b) Each unit shall complete with the base plate, mast, boom, hoist, anchor bolts, provision for moving the hoist, lifting the load and rotating the boom as required to make it as a complete functional system. The hoist and, for jib cranes the trolley movement, shall be electrically powered unless specified otherwise.
- (c) All materials shall be properly selected for the stresses to which they will be subjected. Crane assembly shall be fabricated in accordance with latest standard and regulations
- (d) The crane mast, boom, trolley, hoist and other components shall be fabricated using established procedures and welding shall be carried out by qualified welders.
- (e) Lifting Lugs shall be provided for equipment and components weighing ≥ 40 kg.
- (f) All Anchor Bolts shall be Type 316 stainless steel, sized by equipment manufacturer.
- (g) Free-Standing Jib Crane
 - (i) Crane mast shall be base plate mounted. The crane will have full web gusset plates, welded to the mast and base plate for stability and reinforcement. Insert or sleeve mounted cranes shall not be accepted unless otherwise specified in the Particular Specifications and complying with design requirements as stipulated in the WSH Act.
 - (ii) The crane mast shall be of adequate section size for the specified loading conditions.
 - (iii) The mast height should be selected to have enough head room to facilitate lifting of the required equipment.
 - (iv) The material of construction shall be carbon steel or any other suitable material of manufacturers' standard.
 - (v) One end of the boom should be attached to the mast and should be able to rotate about this connection point.
 - (vi) The top end of the mast shall be finished to attach the crane boom.
- (h) All Jib Cranes
 - (i) The boom shall consist of a standard I-beam or wide flange and may be reinforced with a cap channel for lateral stability.
 - (ii) Trolley stops shall be utilised to prevent the hoist trolley from falling off the boom or contacting any obstruction.
 - (iii) The boom shall be level and horizontal, and shall permit a hoist and trolley to ride upon its flat flange track.
 - (iv) The crane's deflection at the end of the boom shall not exceed 1/150 of the crane's span.
 - (v) Bearings shall be heavy-duty roller type, pre-lubricated and sealed. The bearing shall have a L-10 life of 50,000 hours as minimum.

- (vi) End stops safety features including safety protection to prevent clashing of boom onto side wall, structure and other equipment by limiting swaying of the boom.
- (i) All apparatus supplied under this Section shall be constructed with due regard to safety of operation, accessibility, interchangeability and durability of parts.

G4.3.8 Monorail Hoists Block and Chain Block

- (a) Monorail hoist shall comprise an electric hoist and a rolled steel joist.
- (b) The load chain shall operate the hoist via reduction gear driven wheel. The hoist shall be complete with travel trolley with pull chain, hoisting block, sheaves, hook and hoist brake.
- (c) Chain shall be non-jamming alloy type suitable for one man manual operation with reasonable ease from floor level. Hand wheel shall have a guard that prevents chain slipping or jumping. Chain hoists shall have chain storage adequate for storing full lift length of chain and shall be designed and located to avoid interference while hoisting.
- (d) The hook shall be constructed with sufficient ductility to open noticeably before hook failure, equipped with safety latch, free to rotate 360 degrees with rated load, and positively held in place with locknuts, collars or other devices
- (e) The brakes shall conform to ANSI HST 1M and 2M and be adjustable to compensate for wear. They shall be positive action, Weston type mechanical load brakes with uniform composition lining and forged steel alloy latch pawl.
- (f) The chain block shall be provided with geared hand travel four wheel trolley suitable for running from a rolled steel joist.
- (g) The lifting chain shall be set quality corrosion resistant steel alloy and of such length to allow a lift to be made from the ground level to loading level without re-slinging.
- (h) Electric Hoist
 - (i) The lifting wire rope of the electric hoists shall be of best plough and of such length to allow a lift to be made at ground floor level without re-slinging. The hoist mechanism shall be controlled from a hold on push button pendant.
 - (ii) Hoists shall be fitted with load limited devices unless otherwise specified.
 - (iii) The pendant shall be suspended independently of the electric cable 1300 mm above ground floor level. A stainless steel pendant holder shall be installed on site. Control supply shall be not more than 60 V fed from a two winding transformer with an earthed centre-tap.
 - (iv) Cables feeding the electric hoist unit and down shop cables shall be flexible cables suspended from carriers sliding on taut wires.
 - (v) The electric drive motors shall be fitted with electro-magnetic brakes to prevent over travel on release of the control button.
 - (vi) Limit switches shall be installed to prevent over travel, over lowering and over coiling of the lift rope. End stops and limit switches shall be provided to arrest the hoist block.

- (i) The lifting equipment shall include a triple leg chain sling with hooks; a suitable wire rope sling and a nylon rope sling. The slings shall be of suitable length and strength to enable a safe lift to be made of all items of equipment intended as indicated in the drawings.

G4.3.9 A-Frame Gantry Crane

- (a) The crane shall consist of a horizontal beam supported by two 'A'-shaped end frames/legs.
- (b) The horizontal beam shall be designed to be fitted with lifting devices such as electric or manual hoist, beam clamp, trolley, etc. suited to hoist the concerned equipment.
- (c) The span of the horizontal beam and vertical space between highest hook height and floor level shall be adequate to achieve the required lift taking account of the concerned equipment dimensions and lifting sling length.
- (d) The A-frame gantry crane shall be with heavy duty castor wheels of which two shall be lockable to facilitate easy manoeuvring and positioning.

G4.3.10 Engine Hoist (Portable Crane)

- (a) The engine hoist shall consist of a manual jib crane. The jib shall be manually raised and lowered using a hydraulic ram. The jib shall be supported from a horizontal frame on a minimum of four (4) wheels.
- (b) The hoist shall be capable of folding up to a compact size for storage when not in use.
- (c) The hoist shall include the following safety features:
 - (i) A safety relief valve to prevent overloading;
 - (ii) Lowering valve, or similar, to control the rate which equipment is lowered.
- (d) The boom shall have a minimum of four (4) different extension positions.

G4.3.11 Crane Access

- (a) Fixed aluminium ladders and catwalks complete with toe boards etc. as required for the full and proper access to the crane and associated appurtenances for maintenance purposes shall be provided as per S.O.'s instructions and statutory requirements.
- (b) An aluminium alloy catwalk shall be provided on the crane, complete with handrails and toe boards to enable ease of maintenance of the crane unless otherwise stated.

G4.3.12 Trolley

- (a) Frame
 - (i) Welded steel, cast steel, or ductile iron construction, or a combination thereof. Construct to control deflection of trolley assembly while transmitting the carrying load to running surface.
- (b) Drive shall consist of chain sprocket mounted on shaft. Chain length shall be to within 5 feet of operating floor level. Drive shaft shall drive the trolley wheels through a gear and pinion or spur gear arrangement.

- (c) Furnish roller assembly stabilizers on single-girder trolley units to prevent tipping during load pickup.
- (d) Wheels
 - (i) Rolled or forged steel, accurately machined and ground to receive inner bearing races. Furnish alloy steel axles. Rotating axles with wheels mounted press fit and keys, or with keys alone. Minimum tread hardness 210 Brinell.
- (e) Drive Gears
 - (i) Helical, spur or herringbone type, rolled or cast steel, with machine cut teeth.
- (f) Bearings
 - (i) Combination radial and thrust type, double row, angular contact ball bearings or single-row tapered roller bearings. Bearings pre-lubricated and sealed, or fitted for pressure lubrication. Locate pressure lubrication fittings for accessibility during maintenance.
- (g) Brakes
 - (i) Suitable for service class and rated torque capacities as specified in ANSI B30.11.

G4.3.13 Testing

- (a) The crane shall be tested in the presence of the S.O. to demonstrate the safe condition of the crane.
- (b) Full Load tests in compliance with OSHA, ANSI, BS standards and WSH Council's Code of Practice on Safe Lifting Operations in the Workplaces by an approved person registered with the Chief Inspector of Factories. The load test plan shall be submitted to S.O. for approval.
- (c) Registration with Chief Inspector of Factories: The Contractor shall register the equipment with the Chief Inspector of Factories, Ministry of Manpower.
- (d) In the event the power supply is not available when the crane has been installed, temporary power supply or a hand operated hoist shall be arranged temporarily.
- (e) After all the equipment have been installed, the entire crane assembly including the hoist chain shall be completely tested for all functions in the presence of the S.O. and a licensed tester. The Contractor will be responsible for obtaining a certified test certificate for the crane, a copy of which shall be submitted to the S.O..
- (f) Test complete assemblies for proper alignment and connection, and quiet operation. Contractor shall provide a proper checklist and other test documents for the whole test assemblies. The checklist shall be verified and acknowledged by the qualified and approved tester.
- (g) In addition to complying with the latest standards and regulations, the full load test shall require that weights equalling the maximum capacity of the bridge cranes and hoists be carried by the units as minimum, while the cranes and hoists are operated as follows:
 - (i) Hoists operated through the entire lifting height for each of the units at various positions of the crane and trolley for each crane, as directed by the S.O.;

- (ii) Cranes operate through the entire length of the crane rails.;
- (iii) Hoist trolleys operate through the entire distance along the crane girder;
- (iv) Stop limits fail simulation and braking tests at minimum and maximum speeds;
- (v) If the equipment fails the test, all necessary rectifications shall be made and the equipment shall be re-tested. If the equipment again fails to meet the test requirements to the satisfaction of the S.O., the Contractor shall be liable to replace the entire equipment and reconduct the test; and
- (h) The Contractor shall supply the test load. The test load shall be removed from site by the Contractor after successful tests have been carried out.

END OF SECTION